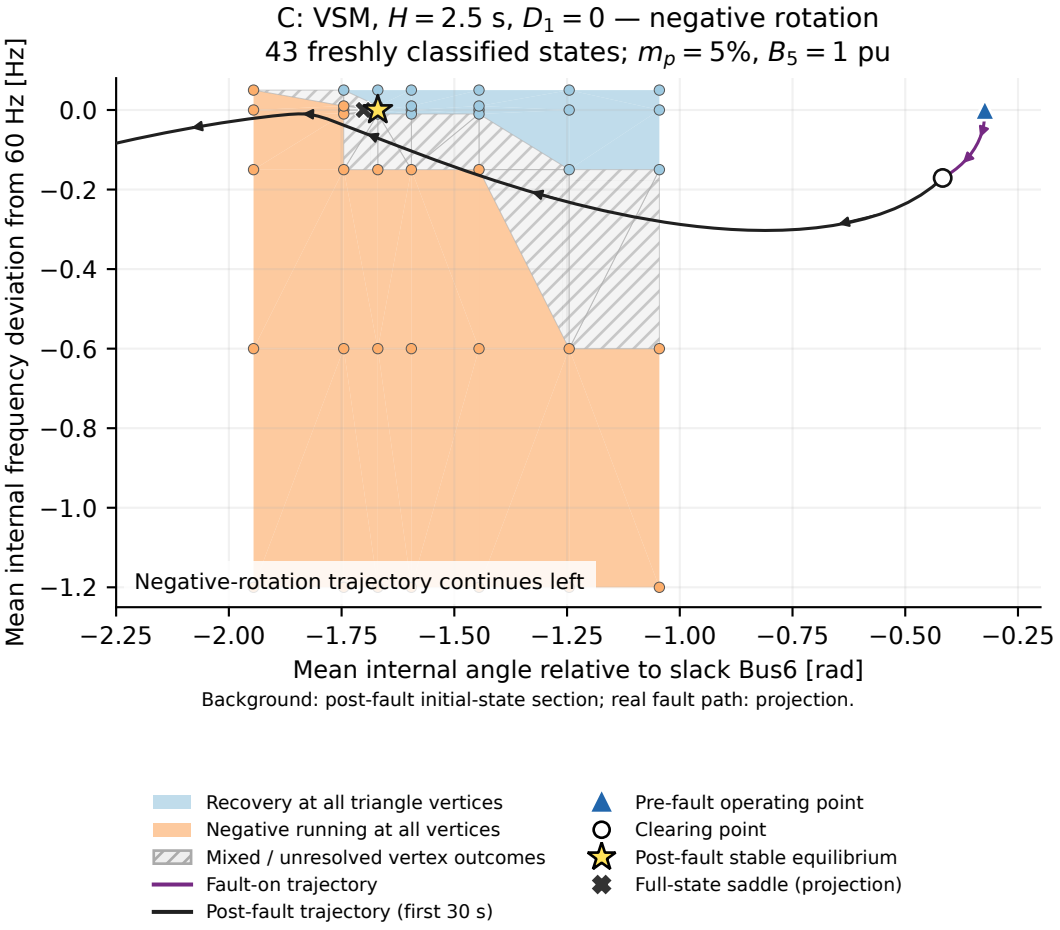
C: VSM, $H = 2.5$ s, $D_1 = 0$ — negative rotation
43 freshly classified states; $m_p = 5\%$, $B_5 = 1$ pu
Mean internal frequency deviation from 60 Hz [Hz]
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
−1.2
Negative-rotation trajectory continues left
−2.25
−2.00
−1.75
−1.50
−1.25
−1.00
−0.75
−0.50
−0.25
Mean internal angle relative to slack Bus6 [rad]
Background: post-fault initial-state section; real fault path: projection.
Recovery at all triangle vertices
Negative running at all vertices
Mixed / unresolved vertex outcomes
Fault-on trajectory
Post-fault trajectory (first 30 s)
Pre-fault operating point
Clearing point
Post-fault stable equilibrium
Full-state saddle (projection)